Muhammad Musa

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Education

Ontario Tech University — Oshawa, Ontario

Bachelor of Engineering, Software Engineering - Highest Distinction (CGPA: 4.12) — *Sep 2022 – May 2026*

- **Relevant Coursework:** Computer Architecture, Systems Programming, Operating Systems, Embedded Systems, Computer Networks, Data Structures, Design & Analysis of Algorithms.

Experience

Programming Data Annotator (LLM Evaluation) — Sep 2024 – Present

Outlier.ai — *Remote*

- Evaluated 1,200+ LLM and agentic tool-use responses in an RLHF pipeline, applying standardized rubrics that raised inter-rater agreement to 95% and fed reward-model training.
- Debugged and corrected code-generation outputs across Python, C++, and 3+ other languages, authoring root-cause reasoning chains and structured feedback that cut recurring code defects in sampled outputs by 30%.

Sensor Data Processing Developer — Mar 2025 – Aug 2025

Autonomous Drive Club, Ontario Tech University — *Oshawa, Ontario*

- Built Python pipelines (NumPy, Open3D) converting raw LiDAR point clouds of 10K+ points per frame into 3D occupancy grids, enabling real-time object detection and environment mapping at 10Hz on an autonomous vehicle.
- Profiled and tuned region-of-interest and height thresholds to cut per-frame compute by 40% with minimal accuracy loss, and built automated regression checks over logged drive data to validate detection.
- Debugged sensor-to-platform integration issues and integrated the perception module with downstream mapping across a 6-member cross-functional team using Agile workflows.

Facilities & Technical Support Assistant — May 2024 – May 2026

Ontario Tech University — *Oshawa, Ontario*

- Diagnosed and resolved hardware, AV, and network faults across campus labs using structured root-cause analysis and ticketing tools, resolving 90%+ of tickets on first contact and escalating complex cases.
- Coordinated daily tasks across a 5-person team and restructured scheduling and workflow, reducing equipment downtime by 25% and delivering operational priorities on time.

Projects

Custom Unix Shell & Process Scheduler | *C, Linux, POSIX syscalls, GDB, Valgrind*

- Built a Unix-style shell in C using low-level POSIX syscalls (fork, exec, wait, pipe, dup2) for process creation, I/O redirection, and pipelines, debugging with GDB and Valgrind to reach zero memory leaks.
- Implemented a multi-policy process scheduler (round-robin, priority, MLFQ) and a multithreaded resource coordinator using mutexes and condition variables, eliminating race conditions verified through automated stress tests.

Microcontroller Firmware & Digital Logic | *C, ARM Cortex-M, GPIO/Registers, Logic Analyzer*

- Wrote bare-metal C firmware interfacing directly with registers, GPIO, and on-chip peripherals (timers, UART, ADC), validating timing and signal behavior on a logic analyzer during board bring-up.
- Designed and tested digital logic for memory, datapaths, and processor organization, verifying correctness against expected waveforms to catch hardware-level faults early.

CiteSeeker | *Python, FastAPI, pgvector, RAGAS, Docker, Claude Code*

- Built an ingestion and hybrid retrieval pipeline (dense embeddings + BM25 with cross-encoder reranking) serving answers at 90%+ citation accuracy while holding p95 latency under 800ms.
- Built a RAGAS evaluation harness with CI checks to catch retrieval regressions automatically, accelerating development with agent-assisted workflows (Claude Code) for scaffolding and debugging.

Technical Skills

Languages: C/C++, Python, Bash, SQL, Perl, Java, JavaScript
Systems, Hardware & Test: Linux, system-level debugging, multithreading & concurrency, computer architecture, embedded & firmware (GPIO/registers), performance & power analysis, test automation, fault injection, GDB
GPU & AI-Native Dev: GPU computing (ROCm/CUDA, OpenCL), agent-based coding workflows (Claude Code, Codex), AI-assisted debugging, RAG, LLM evaluation/RLHF, MCP
Tools: Git, Jira, Agile/Scrum, Docker, PostgreSQL/MySQL, Grafana
Certifications: Google Cloud Gen AI Leader, AWS AI Practitioner, Azure AI Fundamentals, AWS Solutions Architect Associate